

Statistical methods for the cross-cultural recognition memory pipeline

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This note specifies the statistical procedures implemented in **Analysis-Scripts-v2/**. Each section names the function(s) responsible.

1 Data structure

Each participant i in group g contributes one recognition memory session with mixed first-presentation and repeat trials. For each unique stimulus j encountered we record two binary indicators:

- the *hit* indicator $h_{gij} \in \{0, 1\}$, defined on the nonzero-ISI repeat occurrence of stimulus j , equal to `isResponseCorrect`;
- the *false-alarm* indicator $f_{gij} \in \{0, 1\}$, defined on the first (non-repeat) occurrence of stimulus j , equal to $1 - \text{isResponseCorrect}$.

A participant who did not encounter stimulus j contributes NaN at h_{gij} and f_{gij} . The matrices $H_g, F_g \in (\{0, 1\} \cup \{\text{NaN}\})^{n_g \times m}$ are built by `calculateSplitHalfReliability.m`, where n_g is the number of participants in g passing the d' -at-ISI=0 inclusion threshold and m is the union of stimuli encountered by any included participant. Subsequent analyses align to the intersection of stimulus sets across the groups being compared.

2 Within-group split-half reliability

Reliability of an item-level rate vector is estimated by random split-half over participants (default; `splitDim = 1`) or over stimuli (`splitDim = 2`). For each of S random splits ($S = 10,000$ by default), participants are partitioned at random into two halves. Item-level means $\bar{h}^{(1)}, \bar{h}^{(2)} \in \mathbb{R}^m$ are computed within each half and Spearman's rank correlation r_s is computed across items, ignoring items where either half is NaN. Items with fewer than three valid pairs are dropped from that split. The half-set reliability is the *median* of the per-split correlations,

$$\hat{\rho}_{1/2} = \text{median}\{r_s^{(1)}, \dots, r_s^{(S)}\}. \quad (1)$$

The median is used in place of an arithmetic or Fisher- z mean because it is robust to skew and to occasional degenerate splits (e.g., a split where one half's item rates have near-zero variance), and it is invariant to monotone transforms of r . The standard deviation of $\{r_s\}$ is reported as a dispersion statistic.

The reliability of the full set follows from the Spearman–Brown correction,

$$\hat{\rho}_{\text{SB}} = \frac{2\hat{\rho}_{1/2}}{1 + \hat{\rho}_{1/2}}, \quad (2)$$

and is the quantity used for attenuation correction (§5). Implementation: `estimateSplitHalfFlexible.m`. The FA matrix follows the same procedure.

3 Intergroup itemwise correlation: point estimate

Given two groups A and B aligned on a shared item set $\mathcal{S}_{AB}$, define group-level item means

$$\bar{h}_{Aj} = \frac{1}{|\{i : h_{Aij} \neq \text{NaN}\}|} \sum_{i: h_{Aij} \neq \text{NaN}} h_{Aij}, \quad (3)$$

and analogously $\bar{h}_{Bj}$. Item j is retained if it has at least `minResp` non-`NaN` observations in each group (default `minResp` = 2); items below this threshold are removed before correlation. At least `minItems` = 5 retained items are required.

The point estimate is

$$\hat{r}_{AB} = \text{corr}_{\text{Spearman}}(\{\bar{h}_{Aj}\}_{j \in \mathcal{V}_{AB}}, \{\bar{h}_{Bj}\}_{j \in \mathcal{V}_{AB}}), \quad (4)$$

where $\mathcal{V}_{AB} \subseteq \mathcal{S}_{AB}$ are the items meeting the coverage criterion. Spearman is the default given the bounded, often skewed nature of group-mean hit rates.

4 Bootstrap confidence intervals

A cluster bootstrap generates B resampled correlations $\{r_{AB}^{(b)}\}_{b=1}^B$ around the sample point estimate $\hat{r}_{AB}$ (Eq. 4). The headline value is always $\hat{r}_{AB}$; the bootstrap is used only for uncertainty quantification. The bootstrap median is reported as a secondary sanity-check and should sit close to $\hat{r}_{AB}$.

Three CI methods are computed from the same bootstrap distribution. They differ in how they cope with the small- n skew common for correlation statistics. All three are reported by default so divergences between them are visible.

4.1 Resampling units

Participant bootstrap (`BootstrapDim` = 1, **default**). On each of B iterations, participants in group A are resampled with replacement to size n_A , independently from group B . Item-level means $\bar{h}_A^{(b)}, \bar{h}_B^{(b)}$ are recomputed on the resampled participants, the coverage filter is re-applied, and a per-iteration correlation $r_{AB}^{(b)}$ is recorded. Captures uncertainty in “which participants we happened to recruit.”

Stimulus bootstrap (`BootstrapDim` = 2). A single set of stimuli is resampled with replacement from $\mathcal{S}_{AB}$ and used for both groups. Repeatedly drawn stimuli appear as repeated entries in the correlation vector — the standard cluster-bootstrap-by-items quirk. Captures uncertainty in “which sounds we happened to use.”

Reporting both side by side localizes the source of variance; a substantial gap is itself a substantive finding.

4.2 Percentile CI (default)

The simplest summary of the bootstrap distribution:

$$\text{CI}_{1-\alpha}^{\text{pct}} = [\text{P}_{\alpha/2}(\{r^{(b)}\}), \text{P}_{1-\alpha/2}(\{r^{(b)}\})]. \quad (5)$$

Exact when the bootstrap distribution is unbiased and symmetric on the chosen scale. At small n and for correlations near the boundaries, neither assumption holds: the sampling distribution of r is skewed and the bootstrap inherits that skew, so the percentile CI can be biased — typically too narrow on the side closer to ± 1 .

4.3 Fisher- z back-transformed CI

Fisher’s z -transform $z = \text{atanh}(r) = \frac{1}{2} \ln(\frac{1+r}{1-r})$ maps $r \in (-1, 1)$ to $z \in \mathbb{R}$ in a way that makes the sampling distribution of z approximately normal with variance independent of the true ρ . We build a normal-approximation CI on the z scale, then transform the endpoints back:

$$\hat{z} = \text{atanh}(\hat{r}_{AB}), \quad \hat{\sigma}_z = \text{SD}(\{\text{atanh}(r^{(b)})\}), \quad (6)$$

$$\text{CI}_{1-\alpha}^{\text{Fz}} = [\tanh(\hat{z} - z_{1-\alpha/2} \hat{\sigma}_z), \tanh(\hat{z} + z_{1-\alpha/2} \hat{\sigma}_z)], \quad (7)$$

where $z_{1-\alpha/2} = 1.96$ for $\alpha = 0.05$. The interval is symmetric in z but asymmetric in r , with more room on the side away from ± 1 .

Two notes. (i) This is *not* the same as “Fisher- z averaging” of the point estimate, which we explicitly reject. The point estimate remains the untransformed sample $\hat{r}$; only the SE used to build the CI lives in z -space. (ii) Taking percentiles in z -space and back-transforming would give endpoints identical to the raw percentile CI (because $\tanh$ is monotonic), so the meaningful Fisher- z CI must use a parameter such as the bootstrap z -space SD as we do here.

4.4 BCa (bias-corrected and accelerated) CI

When the bootstrap distribution is both biased and skewed — the small- n rule rather than the exception for correlations — Efron’s BCa CI adjusts the percentile quantiles to compensate. Two corrections are estimated from the data.

Bias correction $\hat{z}_0$.

$$\hat{z}_0 = \Phi^{-1}(\widehat{\text{Pr}}(r^{(b)} < \hat{r}_{AB})), \quad (8)$$

where Φ^{-1} is the inverse standard-normal CDF. If the bootstrap distribution is unbiased, exactly half the draws fall below $\hat{r}$ and $\hat{z}_0 = 0$.

Acceleration $\hat{a}$. Estimated from jackknife pseudo-values. Let $r_{(-i)}$ denote the correlation computed with the i -th resampling unit removed (jackknife at the same level as the bootstrap — participants for participant bootstrap, stimuli for stimulus bootstrap), and let $\bar{r}_{(\cdot)}$ be the mean of those n leave-one-out correlations. Then

$$\hat{a} = \frac{\sum_i (\bar{r}_{(\cdot)} - r_{(-i)})^3}{6(\sum_i (\bar{r}_{(\cdot)} - r_{(-i)})^2)^{3/2}}. \quad (9)$$

If the SE of r is constant in r , $\hat{a} = 0$ and BCa reduces to a bias-corrected interval; if the SE varies strongly with r (typical near ± 1), $\hat{a} \neq 0$ further widens or shifts the interval.

Adjusted quantiles. With $z_{\alpha/2}$ and $z_{1-\alpha/2}$ the standard-normal quantiles,

$$\alpha_1 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_{\alpha/2}}{1 - \hat{a}(\hat{z}_0 + z_{\alpha/2})}\right), \quad (10)$$

$$\alpha_2 = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_{1-\alpha/2}}{1 - \hat{a}(\hat{z}_0 + z_{1-\alpha/2})}\right), \quad (11)$$

and

$$\text{CI}_{1-\alpha}^{\text{BCa}} = [\text{P}_{\alpha_1}(\{r^{(b)}\}), \text{P}_{\alpha_2}(\{r^{(b)}\})]. \quad (12)$$

When $\hat{z}_0 = \hat{a} = 0$ this reduces to the percentile CI; otherwise the quantiles shift to correct for bias and skew. BCa is the recommended CI in most modern bootstrap references and is the one to report when the percentile and Fisher- z CIs disagree.

4.5 What to report

Method	Strengths	Failure mode
Percentile (§4.2)	Simple, no parametric assumption	Biased under bootstrap skew
Fisher- z (§4.3)	Symmetric on variance-stabilized scale	Assumes z -space normality
BCa (§4.4)	Second-order accurate; corrects bias+skew	Sensitive to outlying jackknife values

In practice we recommend reporting BCa as the primary CI, with percentile and Fisher- z available as diagnostics. If all three agree, the percentile CI is sufficient for the paper. If BCa diverges substantially from the percentile CI, the bootstrap is operating in its bias/skew regime and the divergence itself should be flagged.

We considered but did not implement a jackknife pseudo-value CI, because BCa already uses the jackknife in its acceleration term and a separate jackknife CI provides little additional information at the small n we work with.

5 Attenuation correction

Reliability-corrected (“disattenuated”) correlations follow the classical Spearman correction:

$$\hat{r}_{AB}^* = \frac{\hat{r}_{AB}}{\sqrt{\hat{\rho}_{\text{SB}}^A \hat{\rho}_{\text{SB}}^B}}. \quad (13)$$

The denominator is floored at machine epsilon to prevent division by very small reliabilities. **Corrected correlations are not clamped** to $[-1, 1]$: when one or both within-group reliabilities are small relative to the raw intergroup correlation, $\hat{r}^*$ can legitimately exceed ± 1 . This is a property of the formula, not a numerical failure — it indicates that the attenuation-corrected relationship at the group level is stronger than the geometric mean of the within-group reliabilities can support, which is itself a substantive observation about how noisy the within-group item structure is. Clamping would hide that signal. Three modes select which reliability is used inside the bootstrap:

‘fixed’ full-sample $\hat{\rho}_{\text{SB}}^{A,B}$ are used unchanged in every iteration. Fast; conservative.

‘subset’ reliabilities are recomputed once on the items surviving the coverage filter at the point estimate, then held fixed.

‘per-draw’ reliabilities are recomputed on each bootstrap resample, using the same resampled participants/stimuli and the same item subset as that iteration’s correlation. The rigorous choice; propagates resample-level uncertainty into the corrected estimate.

6 Paired-bootstrap comparison of three intergroup correlations

The substantive question—whether the US–San Borja correlation differs from US–Tsimane’, etc.—requires testing equality of correlations that *share a group*. Independent bootstrap CIs for each correlation are insufficient: $\hat{r}_{AB}$ and $\hat{r}_{AC}$ are dependent (both depend on A), and ignoring this dependence overstates the variance of their difference.

We use `pairedBootstrapCompareCorrelations.m`, which preserves dependence by *sharing the bootstrap resample of any group that appears in both correlations being compared*. On iteration b :

1. Resample participants within each group independently: indices $\pi_A^{(b)}, \pi_B^{(b)}, \pi_C^{(b)}$ (participant bootstrap), or draw a single shared item resample $\pi_{\text{stim}}^{(b)}$ (stimulus bootstrap).
2. Compute group means from the resampled units, and obtain all three correlations $r_{AB}^{(b)}, r_{AC}^{(b)}, r_{BC}^{(b)}$ on the items meeting `minResp` in all three groups. Critically, $\bar{h}_A^{(b)}$ entering $r_{AB}^{(b)}$ and $\bar{h}_A^{(b)}$ entering $r_{AC}^{(b)}$ are computed from the *same* $\pi_A^{(b)}$, and analogously for B and C .
3. Form pairwise differences $d_{AB,AC}^{(b)} = r_{AB}^{(b)} - r_{AC}^{(b)}$, etc.

The empirical distributions $\{d^{(b)}\}_{b=1}^B$ are the bootstrap distributions of the differences; their $P_{2.5}, P_{97.5}$ percentiles give 95% CIs.

We emphasize that “paired” here refers to the *shared-group resample*, not matched participants across cultures; participants are not matched across sites.

7 Two-sided p-values

Two p-value definitions are computed and reported.

Recentered-null (default). Let $\bar{d}$ denote the bootstrap mean of $d^{(b)}$ and d^{obs} the observed difference computed from the original sample. The null is constructed by recentering the bootstrap distribution at zero, $\tilde{d}^{(b)} = d^{(b)} - \bar{d}$, and

$$p_{\text{null}} = \widehat{\Pr}(|\tilde{d}| \geq |d^{\text{obs}}|), \quad (14)$$

floored at $1/(B+1)$. This is the bootstrap analogue of a permutation test under $H_0 : \delta = 0$ and is the test we recommend reporting.

Straddle-zero (legacy).

$$p_{\text{strad}} = 2 \min(\widehat{\Pr}(d^{(b)} > 0), \widehat{\Pr}(d^{(b)} < 0)). \quad (15)$$

Asks whether the bootstrap CI of the difference crosses zero. Biased under skew of the bootstrap distribution, particularly at small n . Retained for backward comparability with prior pipeline runs.

When (14) and (15) diverge appreciably, the recentered-null variant is the appropriate one to report; the divergence itself signals skew in the bootstrap of d .

8 Small-sample considerations and validation

With per-group $n \in [15, 35]$ and shared-item counts $m \in [40, 80]$ —typical for cross-cultural fieldwork—both the bootstrap CIs and the paired test operate at the lower edge of their valid range. We address this two ways.

Reporting. Each pipeline run records (i) the per-iteration retained item count after the `minResp` filter, (ii) within-group $\hat{\rho}_{\text{SB}}$, and (iii) the gap between (14) and (15). Any of these falling far outside their typical range signals that the test is operating near its limits and the corresponding p-value should be interpreted with caution.

Calibration via simulation. The script `powerSimulationPairedBootstrap.m` generates synthetic three-group recognition memory data with a specified true correlation structure across groups and per-group sample sizes matching the real study. Under H_0 (all pairwise true correlations equal), it reports the empirical type-I error rate of each pairwise test; under H_1 (one correlation differs by Δ), it reports power. CI coverage for each difference is also reported. Running this once at the actual site sizes provides the operating characteristics that should accompany any reported p-value from the real data.

9 Software and reproducibility

All MATLAB analyses use MATLAB 2018b+. Random splits and bootstrap resamples are drawn with `rng('shuffle')` at the top of each routine; for reproducible runs, set `rng(seed)` once before calling `run_cross_cultural_analysis.m`.

A Python twin lives in `python/` and mirrors MATLAB’s `stats/1:1: split_half.py`, `intergroup_corr.py`, `paired_bootstrap.py`, `power_simulation.py`, with shared helpers in `_utils.py`. The top-level driver is `run_cross_cultural_analysis.py`. The two languages compute the same statistics (including the recentered-null and straddle-zero p-values of §7 and the calibration sim of §8) and produce numerically matching results to within bootstrap sampling error. Python 3.7+, NumPy, SciPy.

A Function-to-equation map

Function	Equation / Step
<code>calculateSplitHalfReliability</code>	§1; calls §2
<code>estimateSplitHalfFlexible</code>	$\hat{\rho}_{1/2}$ in §2
Spearman–Brown (inline)	Eq. (2)
<code>bootstrapIntergroupCorrelationSEM</code>	Eqs. (4), (13); §4, §5
<code>pairedBootstrapCompareCorrelations</code>	§6; Eqs. (14), (15)
<code>plotTripleIntergroupBar_v2</code>	reporting
<code>sensitivityMinResp</code>	sensitivity to <code>minResp</code>
<code>powerSimulationPairedBootstrap</code>	calibration